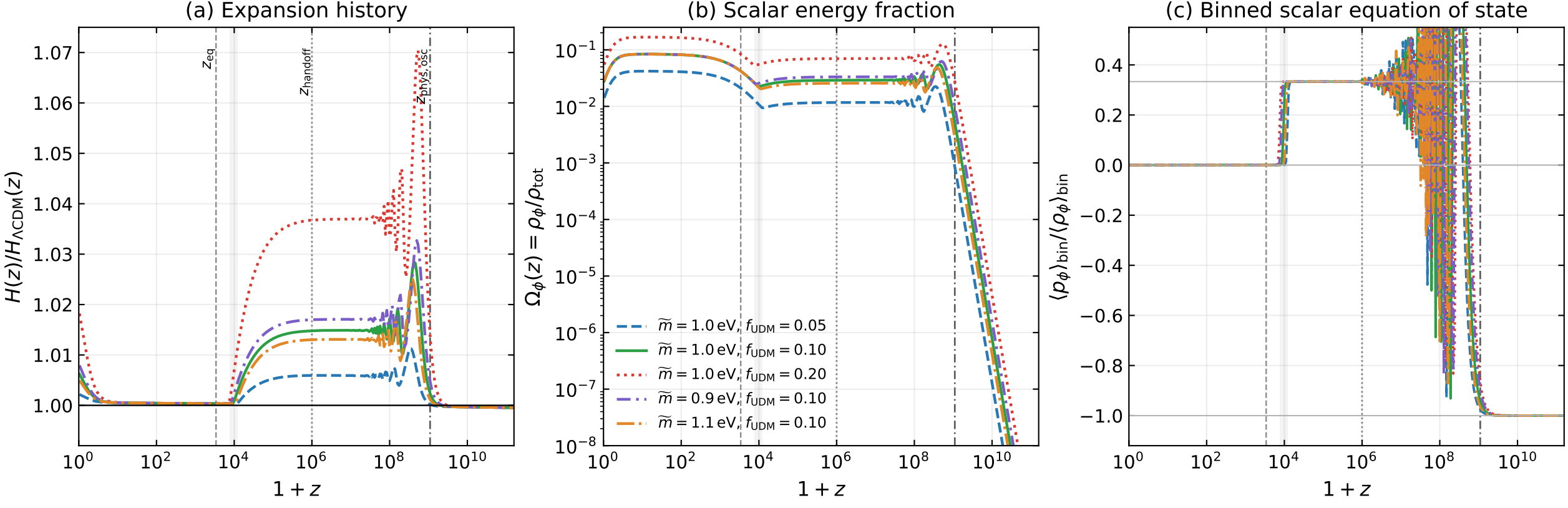
(a) Expansion history
(b) Scalar energy fraction
(c) Binned scalar equation of state
$H(z)/H_{\Lambda\mathrm{CDM}}(z)$
$\Omega_\phi(z) = \rho_\phi/\rho_{\mathrm{tot}}$
$\langle p_\phi\rangle_{\mathrm{bin}}/\langle\rho_\phi\rangle_{\mathrm{bin}}$
$1+z$
z_{eq}
z_{handoff}
$z_{\mathrm{phys,osc}}$
$\widetilde{m} = 1.0\,\mathrm{eV}$, $f_{\mathrm{UDM}} = 0.05$
$\widetilde{m} = 1.0\,\mathrm{eV}$, $f_{\mathrm{UDM}} = 0.10$
$\widetilde{m} = 1.0\,\mathrm{eV}$, $f_{\mathrm{UDM}} = 0.20$
$\widetilde{m} = 0.9\,\mathrm{eV}$, $f_{\mathrm{UDM}} = 0.10$
$\widetilde{m} = 1.1\,\mathrm{eV}$, $f_{\mathrm{UDM}} = 0.10$